

Practice Test — Chapter 4: The Periodic Table

Year 10 Science (Victorian Curriculum, VCSSU123)

Name: _____ Date: _____

Time allowed: 50 minutes Total marks: /50

Instructions: Answer all questions. A periodic table may be used. The answer key is at the end — try the whole test before checking.

Section A — Multiple choice (10 marks, 1 mark each)

Circle the best answer.

1. The modern periodic table arranges elements in order of increasing: - A. atomic mass - B. atomic number - C. density - D. reactivity
 2. The vertical columns of the periodic table are called: - A. periods - B. groups - C. blocks - D. shells
 3. The atomic number of an element tells you the number of: - A. neutrons - B. electron shells - C. protons - D. valence electrons
 4. The maximum number of electrons that can fit in the **second** electron shell is: - A. 2 - B. 8 - C. 18 - D. 32
 5. An element with the electronic configuration **2.8.7** belongs to which group? - A. group 2 - B. group 7 - C. group 17 - D. group 18
 6. When a metal atom loses electrons it forms a: - A. positively charged cation - B. negatively charged anion - C. neutral atom - D. covalent bond
 7. Which group contains the **noble gases**? - A. group 1 - B. group 2 - C. group 17 - D. group 18
 8. As you move **down** group 1 (the alkali metals), reactivity: - A. increases - B. decreases - C. stays the same - D. first increases then decreases
 9. Which type of bond forms when two **non-metals** share electrons? - A. ionic bond - B. covalent bond - C. metallic bond - D. hydrogen bond
 10. Which metal would be extracted from its ore by **electrolysis** rather than by reaction with carbon? - A. copper - B. iron - C. potassium - D. lead
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Section B — Short answer (16 marks)

11. Explain why Mendeleev left **gaps** in his periodic table, and why this made him famous. (3 marks)
12. State **two** physical properties of metals and **two** of non-metals. (4 marks)
13. Write the electronic configuration of: (3 marks)
 - a) magnesium (12 electrons)

- b) oxygen (8 electrons)
- c) sodium (11 electrons)

14. Explain why elements in the **same group** have similar chemical properties. (2 marks)

15. Define each term: (2 marks)

- a) ion
- b) valence electrons

16. Explain why noble gases are **unreactive**. (2 marks)

Section C — Application and problem solving (16 marks)

17. Use the periodic table to identify the element in: (4 marks)

- a) period 3, group 1
- b) period 2, group 17
- c) period 4, group 2
- d) period 1, group 18

18. Complete the table by classifying each element as a **metal**, **non-metal** or **metalloid**. (3 marks)

Element	Classification
Aluminium (Al)	
Silicon (Si)	
Oxygen (O)	

19. Group 1 metals react with water. (4 marks)

- a) Write a **word equation** for sodium reacting with water. (2 marks)
- b) Name the **two** products formed when any alkali metal reacts with water. (2 marks)

20. Use the activity series (most → least reactive: K, Na, Ca, Mg, Al, Zn, Fe, Ni, Sn, Pb, Cu, Ag, Au) to predict whether a reaction will occur. Write “reaction” or “no reaction”. (5 marks)

- a) copper sulfate + zinc
 - b) zinc sulfate + copper
 - c) iron sulfate + sodium
 - d) magnesium sulfate + zinc
 - e) Explain your reasoning for part (a). (1 mark)
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Section D — Extended response (8 marks)

21. Sodium (a group 1 metal) reacts with chlorine (a group 17 non-metal) to form sodium chloride. Explain, in terms of **electrons, ions and bonding**, how this ionic compound forms. Use the words *valence electrons, cation, anion* and *ionic bond*. (5 marks)

22. Compare **ionic** and **covalent** bonding by completing the table. (3 marks)

	Ionic bonding	Covalent bonding
Occurs between...		
Electrons are...		
Relative bond strength		

ANSWER KEY — The Periodic Table

Section A 1. B — atomic number 2. B — groups 3. C — protons 4. B — 8 5. C — group 17 6. A — positively charged cation 7. D — group 18 8. A — increases 9. B — covalent bond 10. C — potassium

Section B

11. He left gaps for elements that had not yet been discovered; he predicted their properties based on the elements around them; when those elements (e.g. germanium) were later discovered with the predicted properties, it confirmed his table. (3)

12. Metals (any 2): shiny, malleable, ductile, good conductors, high melting points, dense. Non-metals (any 2): dull, brittle, poor conductors, low melting/boiling points. (4)

13. a) 2.8.2; b) 2.6; c) 2.8.1. (3)

14. They have the same number of valence (outer-shell) electrons, so they bond and react with other substances in the same way. (2)

15. a) ion = a charged atom formed by losing or gaining electrons; b) valence electrons = the electrons in the outermost shell of an atom. (2)

16. Noble gases have a full outer shell of electrons, so they do not need to gain or lose electrons to become stable; therefore they don't form bonds or ions. (2)

Section C

17. a) sodium (Na); b) fluorine (F); c) calcium (Ca); d) helium (He). (4)

18. Aluminium = metal; Silicon = metalloid; Oxygen = non-metal. (3)

19. a) sodium + water → sodium hydroxide + hydrogen (2) b) Hydrogen (gas) and an alkali / metal hydroxide (sodium hydroxide). (2)

20. a) reaction (zinc is more reactive than copper, so it displaces it); b) no reaction (copper is less reactive than zinc); c) reaction (sodium is more reactive than iron); d) no reaction (zinc is less reactive than magnesium). (4) e) Zinc is higher in the activity series than copper, so the more reactive zinc displaces the less reactive copper from copper sulfate. (1)

Section D

21. Marking points (5): Sodium has **1 valence electron**; chlorine has 7 valence electrons (needs 1 more). Sodium **loses** its valence electron to chlorine. Sodium becomes a positively charged **cation** (Na^+); chlorine becomes a negatively charged **anion** (Cl^-). The oppositely charged ions are attracted by strong electrostatic force, forming an **ionic bond** / ionic compound (NaCl), which arranges into a lattice. (1 mark each, max 5.)

22. (3 — 1 mark per row roughly)

	Ionic bonding	Covalent bonding
Occurs between...	a metal and a non-metal	two non-metals
Electrons are...	transferred / donated	shared
Relative bond strength	strong	strong

END OF TEST